

PAPER_680: Vortex Quantization in the UQFF Aether Superfluid

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Framework: Unified Quantum Field Framework (UQFF) v5.30

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Class: VortexQuantization | CP4 #264

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Domain: Quantum Fluids / Superfluidity

Abstract

Quantized vortices arise naturally in the UQFF Aether superfluid when angular momentum exceeds $n\hbar$. We compute the circulation $\kappa_v = nh/m_{\text{UA}}$, vortex core $a_v \approx \xi_{\text{UA}}e^{-n\pi}$, and vortex energy per unit length $E_v/L = \rho_{\text{UA}}\kappa_v^2/(4\pi) \ln(R/a_v) \cdot (\rho_{\text{UA}}/\rho_{\text{SCm}})$. The UQFF Magnus force on a vortex is enhanced by $(1 + f_{\text{TRZ}}\rho_{\text{UA}}/\rho_{\text{SCm}})$.

Primary UQFF Equation

$$\kappa_v = \frac{nh}{m_{\text{UA}}}, \quad \frac{E_v}{L} = \frac{\rho_{\text{UA}}\kappa_v^2}{4\pi} \ln \frac{R}{a_v} \cdot \frac{\rho_{\text{UA}}}{\rho_{\text{SCm}}}$$

Parameters

Parameter	Value	Description
n	1, 2, 3...	Winding number
R	system radius	Outer boundary
a_v	$\xi e^{-n\pi}$	Vortex core radius

UQFF Constants

Constant	Value	Description
ρ_{UA}	$7.09 \times 10^{-36} \text{ kg/m}^3$	Universal Aether density
ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	Schwarzschild condensate
f_{TRZ}	0.1	Time-reversal zone factor
κ	$5 \times 10^{-4} \text{ day}^{-1}$	UQFF calibration constant
[SSq]	0.57	Superstring quenching factor
μ_J	$3.38 \times 10^{23} \text{ J}\cdot\text{m}$	Magnetic string coupling
γ	$5 \times 10^{-5} / 86400 \text{ s}^{-1}$	Aether oscillation decay

C++ Implementation

```
#include "VortexQuantization.h"
```

```
VortexQuantization obj;  
auto result = obj.compute_primary(...); // primary equation  
obj.simulate_over_M(M_start, M_end, dM, "output.csv");
```

Python CP4 Calculator

```
from CondensedPhysics2 import VortexQuantizationCalculator
```

```
calc = VortexQuantizationCalculator()  
result = calc.compute() # returns all UQFF quantities  
print(result)
```

References

Feynman 1955 (quantized vortices); Abo-Shaeer et al. 2001 (BEC vortices); UQFF Session 173

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